Machine Learning Project Report

Dual-Pipeline Analysis: Regression & Classification

Introduction

This document outlines the complete machine learning pipeline implemented in **miniproject.ipynb**. The project features a dual-pipeline approach, demonstrating continuous regression on the Diamonds dataset and multinomial classification on the Water Potability dataset. It explains **what** was done, **where** it was implemented, and **why** specific methodologies were chosen.

Part 1: Continuous Regression Pipeline

Dataset: Diamonds

What is done:

Predicting the continuous numerical price of diamonds based on their physical characteristics.

Where is it done:

Steps 1.1 through 1.7 in miniproject.ipynb.

Why is it done:

To demonstrate linear modeling techniques and feature impact analysis on a continuous target variable.

Detailed Pipeline Execution:

1. Data Loading (Step 1.1): Retrieved the dataset via kagglehub.

2. Preprocessing & Data Splitting (Step 1.2): Separated features and target. Used One-Hot Encoding to convert categorical data ('cut', 'color', 'clarity') into numeric forms for model compatibility. Split data into 80% training and 20% testing sets to evaluate unseen performance. Applied StandardScaler to continuous features ensuring uniform gradient descent and preventing features with larger scales from dominating the distance metrics.

3. Model Training (Steps 1.3 - 1.5): Trained three distinct linear models: Standard Linear Regression, Lasso Regression (L1 regularization to force sparsity and perform feature selection), and Ridge Regression (L2 regularization to prevent extreme weights and overfitting).

4. Evaluation & Visualization (Step 1.6): Compared models using R^2 Score and Mean Squared Error (MSE), plotted with Seaborn to provide visual clarity on model performance.

5. Feature Importance (Step 1.7): Extracted coefficients from the Lasso model to quantitatively demonstrate which physical features of a diamond most heavily dictate its price.

Part 2: Multinomial Classification Pipeline

Dataset: Water Potability

What is done:

Classifying water samples into three discrete potability categories based on their chemical attributes.

Where is it done:

Steps 2.1 through 2.11 in miniproject.ipynb.

Why is it done:

To implement robust preprocessing that prevents data leakage, resolve class imbalances using synthetic resampling, and evaluate advanced non-linear classification models.

Detailed Pipeline Execution:

1. Data Loading & Target Engineering (Step 2.1): Downloaded data. Since the goal was multinomial classification, a synthetic 3-class target ('Potability_Class') was engineered based on pH quantiles.

2. Preprocessing & Anti-Leakage (Step 2.2): The original 'ph' column was explicitly dropped to prevent severe data leakage. Handled missing data by imputing median values to maintain dataset integrity. Scaled the data with StandardScaler.

3. SMOTE Resampling (Step 2.2): Applied the Synthetic Minority Over-sampling Technique (SMOTE) strictly on the training set. This was crucial to resolve class imbalance and ensure the models wouldn't become biased towards the majority class.

4. Model Initialization & Training (Steps 2.3 - 2.9): Trained an array of algorithms ranging from simple to complex: Logistic Regression (L1 & L2 solvers), Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost.

5. Evaluation Matrices (Steps 2.4 - 2.9): Calculated Accuracy, Precision, Recall, and Weighted F1-Scores. Visualized detailed performance breakdowns via Confusion Matrices.

6. Final Visual Comparison & Insights (Steps 2.10 - 2.11): Compared all classifiers based on their F1-Scores on a seaborn bar plot. Finally, extracted feature importances via the Random Forest model to identify the most critical chemical components determining water safety.

Conclusion:

The pipeline demonstrates an end-to-end best-practice workflow for ML: proper data isolation, anti-leakage principles, rigorous feature scaling, imbalance resolution via SMOTE, and comprehensive metric evaluation across varying model complexities.